Component 6:
Intro to Sequential Circuits

- **Introduction to sequential circuits**
- **Types of sequential circuits**
- **Storage elements**
  - **Latches**
  - **Flip-flops**
- **Sequential circuit = Finite State Machine**

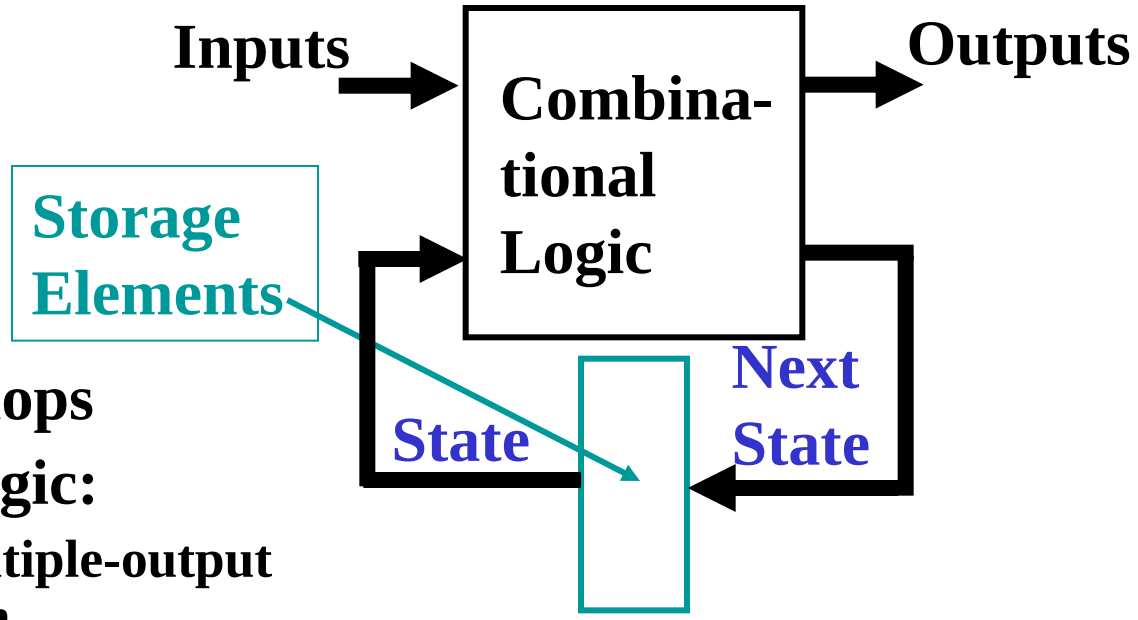
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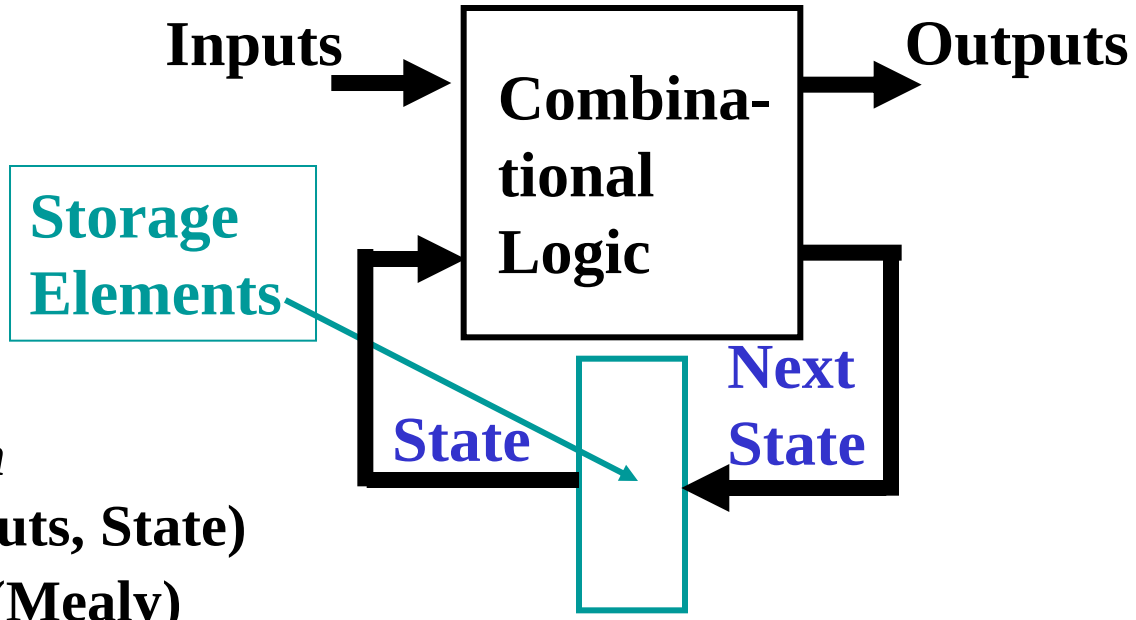
Introduction to Sequential Circuits



- **A Sequential circuit contains:**
  - **Storage elements: Latches or Flip-Flops**
  - **Combinatorial Logic:**
    - **Implements a multiple-output switching function**
    - **Inputs are signals from the outside.**
    - **Outputs are signals to the outside.**
    - **Other inputs, State or Present State, are signals from storage elements.**
    - **The remaining outputs, Next State are inputs to storage elements.**

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Introduction to Sequential Circuits



- Combinatorial Logic
  - *Next state function*
    Next State = f(Inputs, State)
  - *Output function* (Mealy)
    Outputs = g(Inputs, State)
  - *Output function* (Moore)
    Outputs = h(State)
- Output function type depends on specification and affects the design significantly

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Types of Sequential Circuits

- **Depends on the times at which:**
  - **storage elements observe their inputs, and**
  - **storage elements change their state**
- **Synchronous**
  - **Behavior defined from knowledge of its signals at discrete instances of time**
  - **Storage elements observe inputs and can change state only in relation to a timing signal (clock pulses from a clock)**
- **Asynchronous**
  - **Behavior defined from knowledge of inputs an any instant of time and the order in continuous time in which inputs change**

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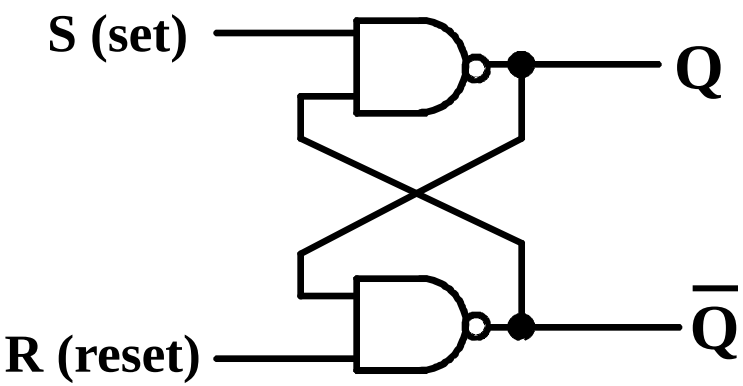
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Basic (NAND) $\bar{S}$ – $\bar{R}$ Latch

- "Cross-Coupling" two NAND gates gives the $\bar{S}$ -$\bar{R}$ Latch:
- Which has the time sequence behavior:



Time

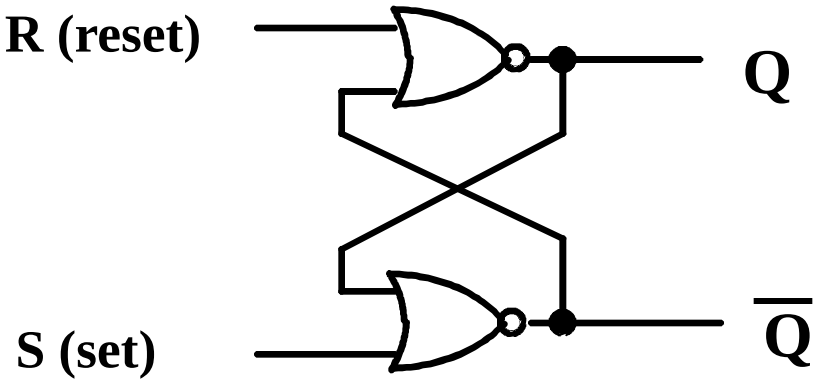
R	S	Q	$\bar{Q}$	Comment
1	1	?	?	Stored state unknown
1	0	1	0	"Set" Q to 1
1	1	1	0	Now Q "remembers" 1
0	1	0	1	"Reset" Q to 0
1	1	0	1	Now Q "remembers" 0
0	0	1	1	Both go high
1	1	?	?	Unstable!

- S = 0, R = 0 is <u>forbidden</u> as input pattern

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Basic (NOR) S – R Latch

- **Cross-coupling two NOR gates gives the S – R Latch:**
- **Which has the time sequence behavior:**



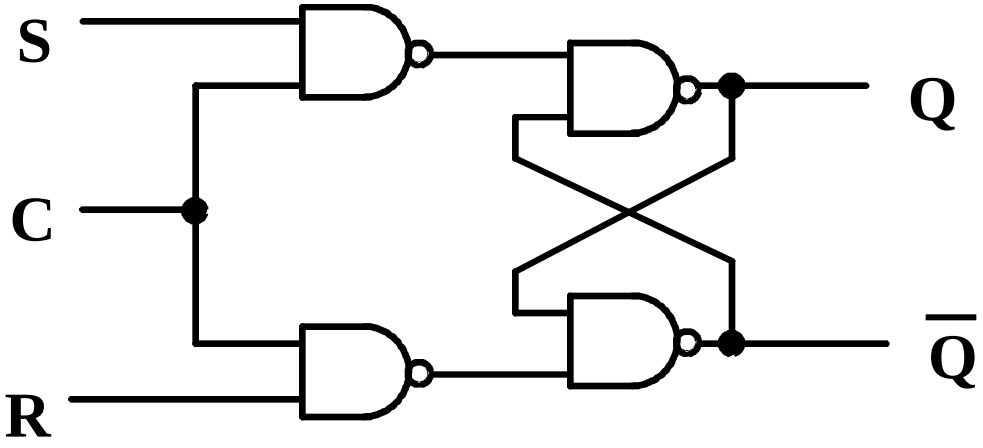
Time

R	S	Q	$\overline{Q}$	Comment
0	0	?	?	Stored state unknown
0	1	1	0	"Set" Q to 1
0	0	1	0	Now Q "remembers" 1
1	0	0	1	"Reset" Q to 0
0	0	0	1	Now Q "remembers" 0
1	1	0	0	Both go low
0	0	?	?	Unstable!

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Clocked S - R Latch

- **Adding two NAND gates to the basic $\overline{S}$ - $\overline{R}$ NAND latch gives the clocked S – R latch:**

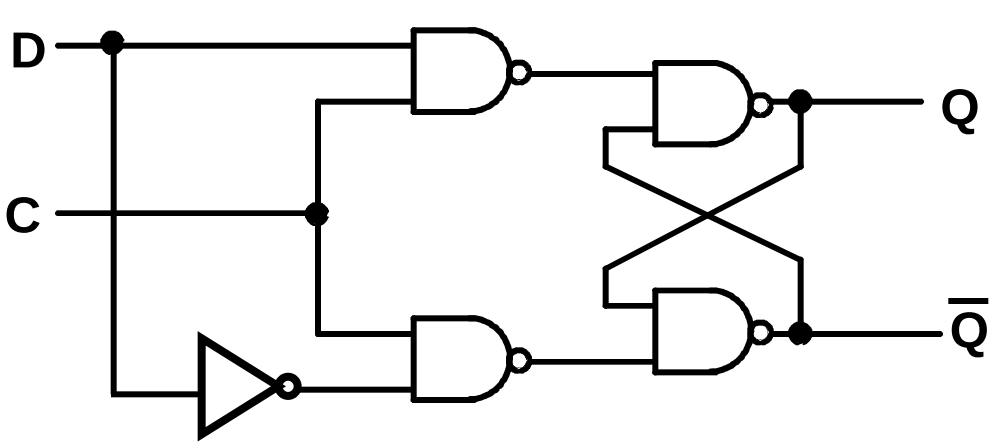


- **Has a time sequence behavior similar to the basic S-R latch except that the S and R inputs are only observed when the line C is high.**
- **C means "control" or "clock".**

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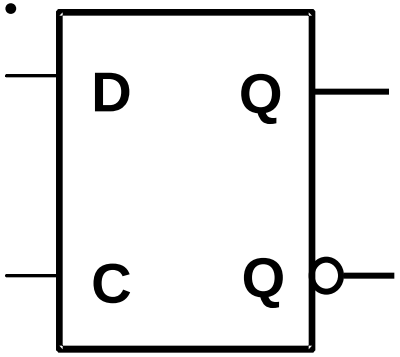
D Latch

- Adding an inverter to the S-R Latch, gives the D Latch:
- Note that there are no "indeterminate" states!



Q	D	Q(t+1)	Comment
0	0	0	No change
0	1	1	Set Q
1	0	0	Clear Q
1	1	1	No Change

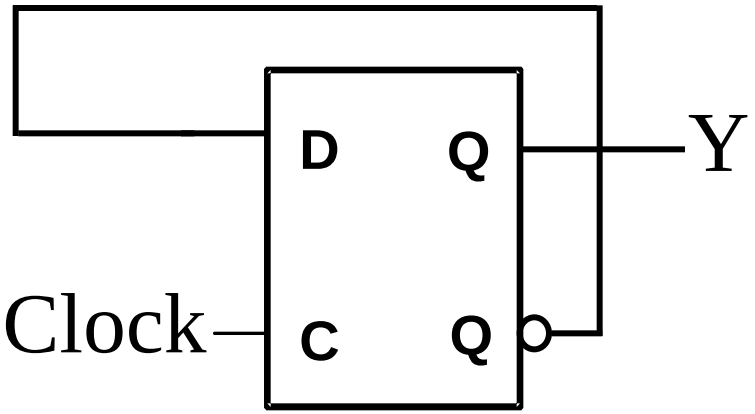
The graphic symbol for a D Latch is:



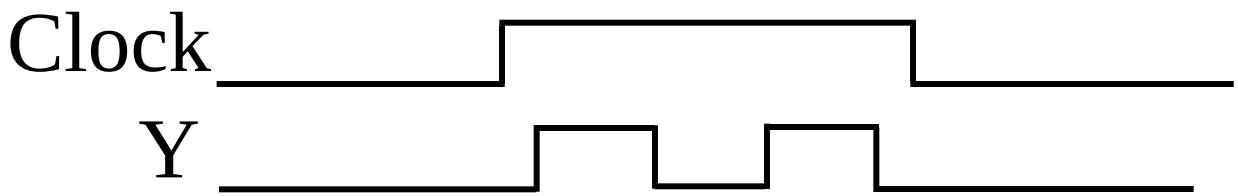
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The Latch Timing Problem

- **Consider the following circuit:**



- **Suppose that initially Y = 0.**



- **As long as Clk = 1, the value of Y continues to change!**
  - **Oscillation!**
- **<u>Desired behavior</u>: Y changes <u>only once</u> per clock cycle**

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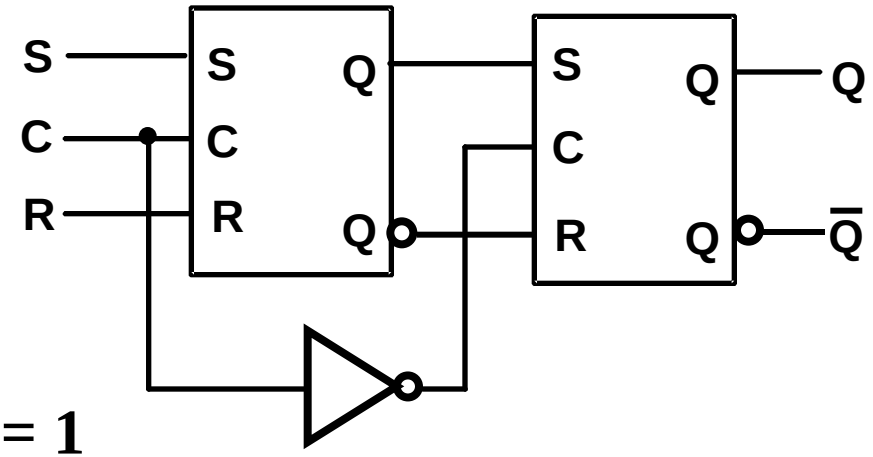
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S-R Master-Slave Flip-Flop

- Consists of two clocked S-R latches in series with the clock on the second latch inverted



- The input is observed by the first latch with C = 1
- The output is changed by the second latch with C = 0
- The path from input to output is broken by the difference in clocking values (C = 1 and C = 0).
- The behavior demonstrated by the example with D driven by Y given previously is prevented since the clock must change from 1 to 0 before a change in Y based on D can occur.

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Flip-Flop Solution

- **Use edge-triggering instead of master-slave**
- **An *edge-triggered* flip-flop ignores the pulse while it is at a constant level and triggers only during a transition of the clock signal**
- **Edge-triggered flip-flops can be built directly at the electronic circuit level, or**
- **A master-slave D flip-flop which also exhibits edge-triggered behavior can be used.**

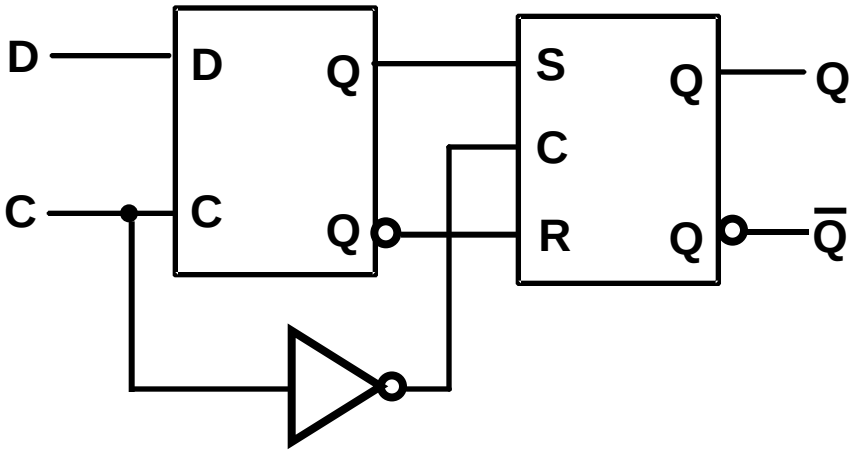
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Edge-Triggered D Flip-Flop

- The edge-triggered D flip-flop

  =

  the master-slave D flip-flop



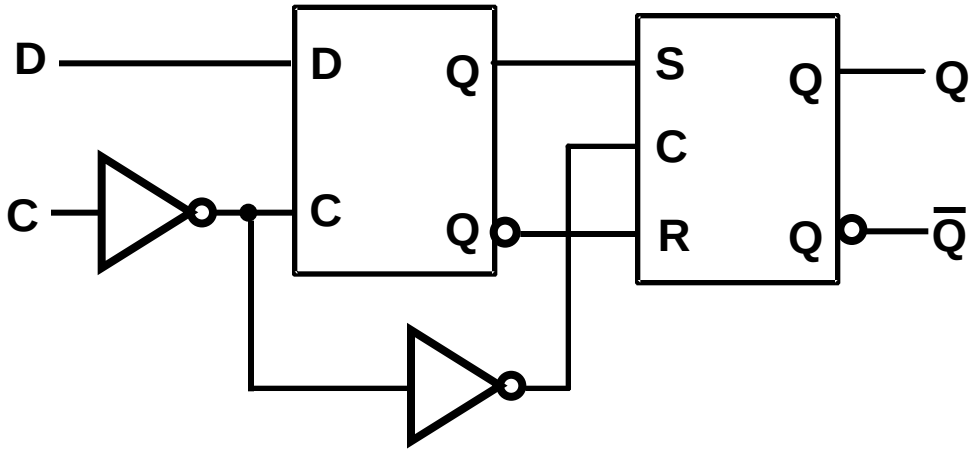
- The change of the D flip-flop output is happens **at the negative edge** (high-to-low) of clock

- It is called a *negative-edge triggered* flip-flop

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Positive-Edge Triggered D Flip-Flop

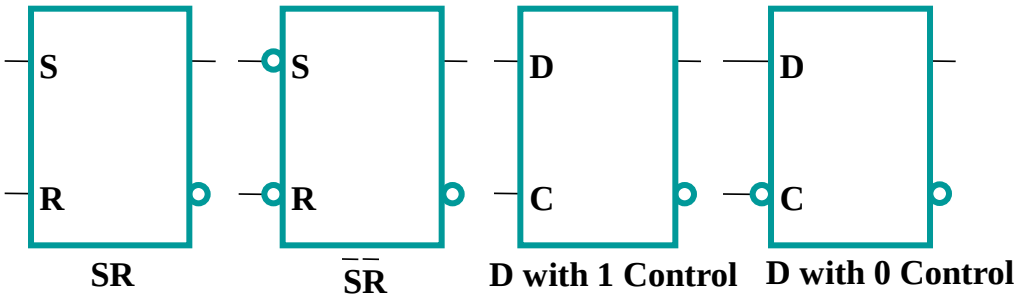
- **Formed by adding inverter to clock input**



- **The change of the D flip-flop output is happens at the positive edge (low-to-high) of clock**
- **It is called a *positive-edge triggered* flip-flop**
- **Commonly used**

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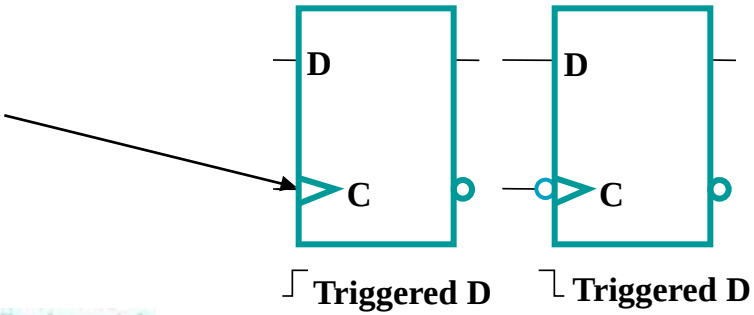
Standard Symbols for Storage Elements



(a) Latches

Edge-Triggered:

Dynamic indicator

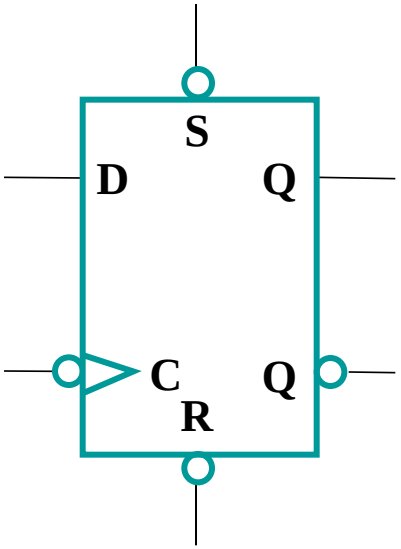


(b) Edge-Triggered Flip-Flops

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Direct (Async) Inputs

- At power up or at reset, all or part of a sequential circuit usually is initialized to a known state before it begins operation
- This initialization is often done outside of the clocked behavior of the circuit, i.e., asynchronously.



- Direct R and/or S inputs that control the state of the latches within the flip-flops are used for this initialization.
- For the example flip-flop shown
  - 0 applied to $\overline{R}$ resets the flip-flop to the 0 state
  - 0 applied to $\overline{S}$ sets the flip-flop to the 1 state

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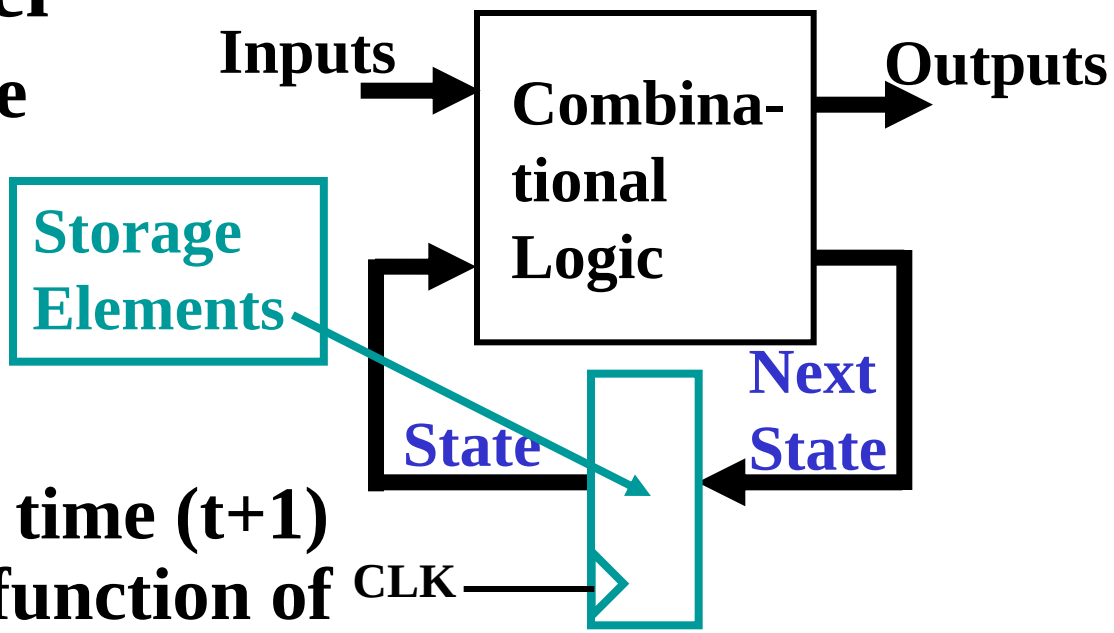
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Sequential Circuit Analysis

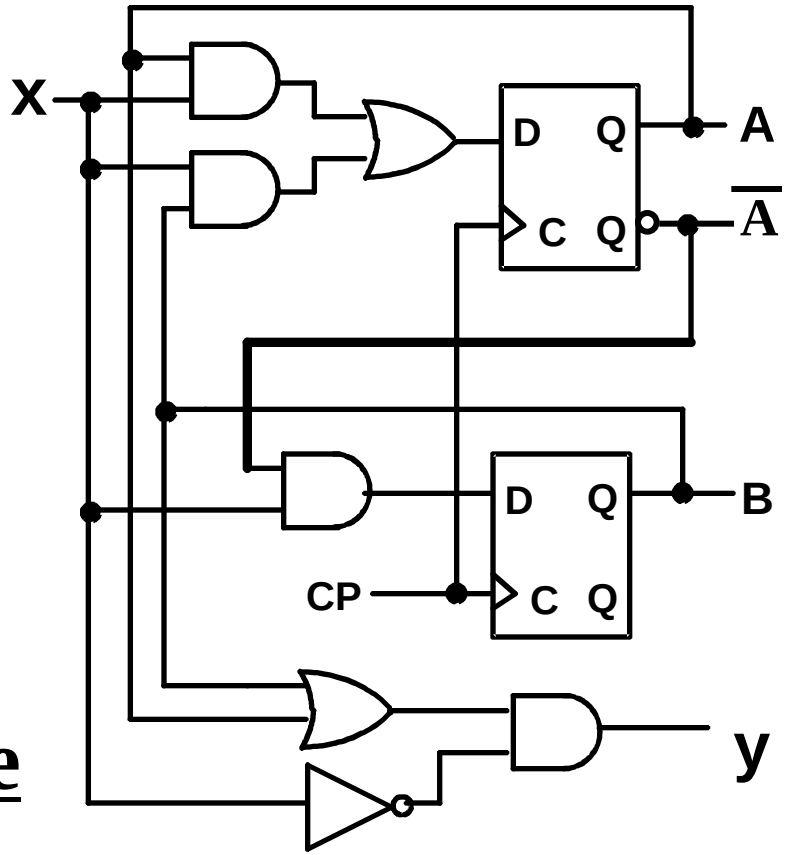
- **General Model**
  - **Current State at time (t) is stored in an array of flip-flops.**
  - **Next State at time (t+1) is a Boolean function of State and Inputs.**
  - **Outputs at time (t) are a Boolean function of State (t) and (sometimes) Inputs (t).**



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Example

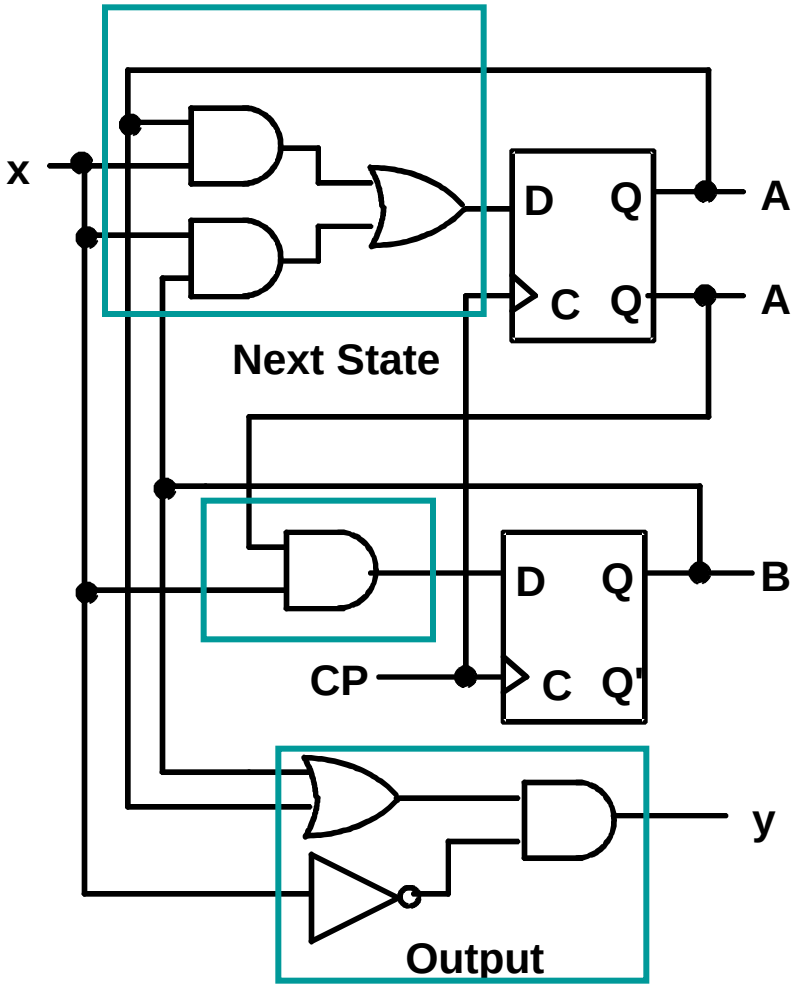
- **Input:** x(t)
- **Output:** y(t)
- **State:** (A(t), B(t))
- **What is the Output Function?**
- **What is the Next State Function?**



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Example (continued)

- **Boolean equations for the functions:**
  - **A(t+1) = A(t)x(t) + B(t)x(t)**
  - **B(t+1) = $\overline{A}$(t)x(t)**
  - **y(t) = $\overline{x}$(t)(B(t) + A(t))**



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State Table Characteristics

- ***State table*** **– a multiple variable table with the following four sections:**
  - ***Present State*** **– the values of the state variables for each allowed state.**
  - ***Input*** **– the input combinations allowed.**
  - ***Next-state*** **– the value of the state at time (t+1) based on the present state and the input.**
  - ***Output*** **– the value of the output as a function of the present state and (sometimes) the input.**
- **From the viewpoint of a truth table:**
  - **The inputs are Input, Present State**
  - **The outputs are Output, Next State**

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Example: State Table

- **The state table can be filled in using the next state and output equations:** $A(t+1) = A(t)x(t)+B(t)x(t)$

$$B(t+1) = \overline{A}(t)x(t)$$

$$y(t) = \overline{x}(t)(B(t) + A(t))$$

Present State		Input	Next State		Output
A(t)	B(t)	x(t)	A(t+1)	B(t+1)	y(t)
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

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State Diagrams

- **The sequential circuit function can be represented in graphical form as a state diagram with the following components:**
  - **A circle with the state name in it for each state**
  - **A directed arc from the Present State to the Next State for each state transition**
  - **A label on each directed arc with the Input values which causes the state transition, and**
  - **A label:**
    - **On each circle with the output value produced, or**
    - **On each directed arc with the output value produced.**

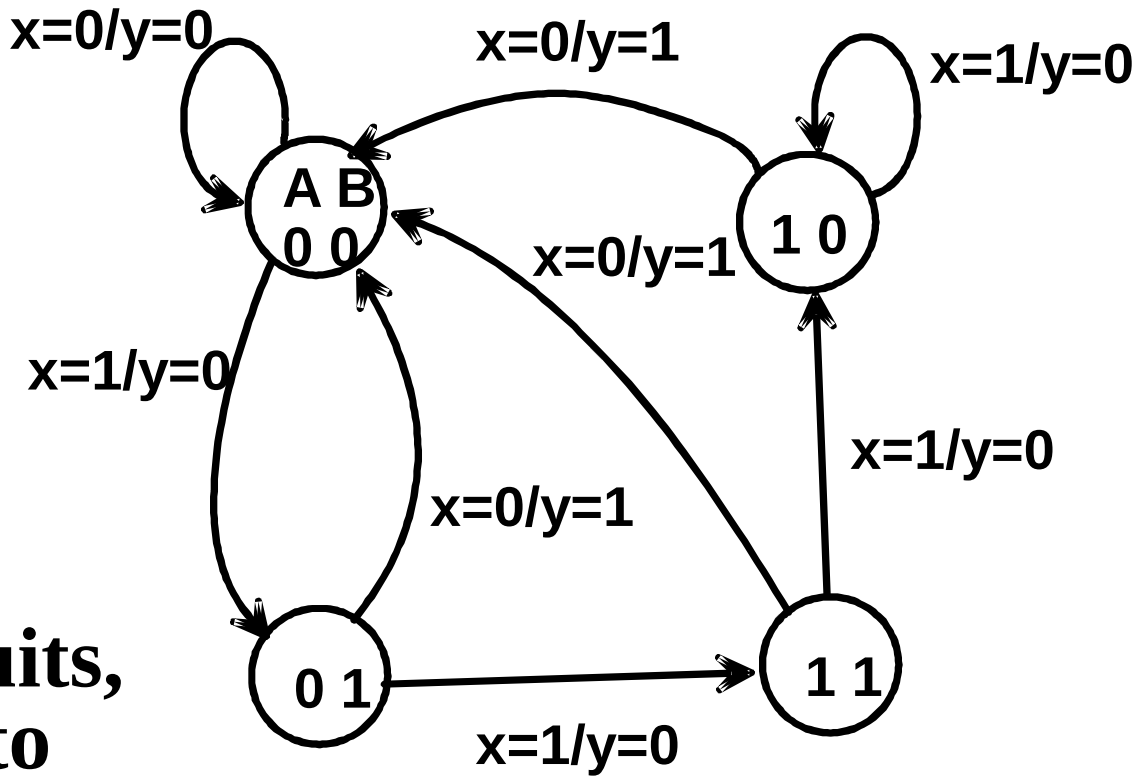
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State Diagrams

- **Label form:**
  - **On circle with output included:**
    - **state/output**
    - **Moore type output depends only on state**
  - **On directed arc with the output included:**
    - **input/output**
    - **Mealy type output depends on state and input**

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Example: State Diagram



- **Which type?**
- **For small circuits, usually easier to understand than the state table**

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